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Trajectory Planning for UAV Based on Improved ACO Algorithm

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ABSTRACT Trajectory planning is an important subject in the field of unmanned aerial vehicles (UAVs). However, existing methods do not solve some problems well, such as slow convergence speed and low searching efficiency of related algorithms and collisions between UAVs and the obstacles. Therefore, a method is proposed to solve a trajectory planning problem for multi-UAV in a static environment, which includes three main phases: the initial trajectory generation, the trajectory correction, and the smooth trajectory planning. First, the improved algorithm called MACO which introduces the metropolis criterion into the node screening mechanism of the ant colony optimization (ACO) algorithm is presented to generate the initial trajectory that can effectively avoid falling into the local optimal solution and stagnation. Then, considering size constraint of UAVs, this paper gives three trajectory correction schemes to solve the collision avoidance problem and further to optimize the initial trajectory. Last, the discontinuity originated from the sharp turn which occurred in the trajectory planning is solved by the inscribed circle (IC) smooth method, and the produced trajectory has shown better performance in reducing fuel consumption and improving the trajectory safety. Experimental results demonstrate that the proposed method has the high feasibility and effectiveness from aspects of the optimal solution, the collision avoidance, and the smooth trajectory in the trajectory planning problem for UAVs.

INDEX TERMS Trajectory planning, unmanned aerial vehicles, trajectory correction, metropolis criterion, IC smooth.

I. INTRODUCTION

Unmanned aerial vehicle (UAV) is an unmanned aircraft that can autonomously complete a given task [1], such as monitoring and assessment of natural disaster, search, rescue, and surveillance. Due to the advantages of multi-UAV in information sharing, strong robustness, and the ability, it can perform significantly better than the single UAV in the complex tasks. Hence, it has been the subject of the trajectory planning.

Trajectory planning is one of the most important techniques of unmanned aerial vehicle research and it finds a feasible optimal trajectory between the initial control point and target control point [2]. The trajectory planning usually can be divided into the global trajectory planning and local

trajectory planning according to whether the environmental state is known or not. When the environmental state is known and the geographic information is non-dynamic, the global trajectory planning can achieve a safe and feasible trajectory. And it is composed of orderly trajectory nodes from the initial point to the target point. The local trajectory planning can generate higher quality trajectory efficiently in a dynamic environment with unknown environmental conditions and variable terrains.

In recent years, there has been an increasing interest in trajectory planning. The intelligent algorithm and its variant are popular among researchers because the traditional trajectory planning algorithm does not achieve the desired results both in the global trajectory planning and local trajectory planning. Babel [1] presented a method to solve the problem that the UAVs can arrive at the destination simultaneously or

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sequentially with the specified time delays. Gao *et al.* [3] divided the global optimization problem into some local optimization problems for a specific task, and proposed the search-attack mission self-organization algorithm (SAMSOA) to achieve the trajectory planning. In [4], Dubins curve is employed to smoothly connect the waypoints which are generated by the ACO algorithm, so an intelligent self-organized algorithm (ISOA) is presented to cope with the distributed control architecture problem. Wu *et al.* [5] proposed a hybrid particle swarm optimization (PSO) algorithm by introducing the metropolis criterion, and it can accept the suboptimal solution with the certain probability to guarantee that the algorithm can jump out of the local optimum. Moreover, the Rauch-Tung-Striebel (RTS) smoother is improved to smooth the generated trajectory. Collision free 4D trajectory planning for multiple UAVs is addressed by Liu *et al.* [6], and a spatial refined voting mechanism is designed to overcome the shortcomings that traditional PSO algorithm is slow in convergence and easily falls into the local optimum. In addition, a method of time coordination is proposed to generate the specified trajectories for each UAV, and it ensures that the air vehicles can reach the same target with the same time consumption. The problem of trajectory planning usually can be modeled as a nonlinear optimal control problem with non-convex constraints, therefore, an algorithm is proposed to obtain the optimal solution quickly by approximating the non-convex problem to a series of continuous convex programming problems [7]. Goel *et al.* [8] proposed the glow swarm optimization (GSO) algorithm to find the collision-free trajectory in three dimensional dynamic environment. Song *et al.* [9] designed the multimodal delayed particle swarm optimization (MDPSO) algorithm to deal with the problem of the global smooth trajectory planning, and the Bezier curve is applied to seek a feasible and optimal trajectory. To solve the problem of multiple UAVs executing multiple ground targets, a method based on the ACO algorithm and the improved Dubins path is proposed, and the multiple targets problem can be transformed to an optimization problem. The targets are assigned to UAVs by means of the ACO algorithm and the trajectory is generated using the improved Dubins path [10]. For the deadlock problem, the improved algorithm is proposed to find the optimal trajectory with few iterations [11]. In [12], the best set of waypoints can be found by the occlusion aware waypoint generation algorithm for taking pictures in target area, and the trajectories for all UAVs are generated using the improved genetic algorithm. The improved method based on the grey wolf optimization algorithm [13] is presented to find the optimal trajectory under a dynamic environment. In [14], a method of trajectory planning for UAVs is designed to ensure the specified image overlap. In addition, the improved ACO algorithm is used to get the near optimized trajectory. The improved fruit fly optimization algorithm [15] is proposed to find the optimal trajectory of UAVs in the complex three dimensions environment by considering various performance and constraints. In [16], the collision probability

model of UAV is established to effectively solve the problem of trajectory conflicts for multi-UAV within the constraints of the number of aircrafts and search space, and the kalman state estimation method is used to solve the problem of communication inconsistency in a dynamic environment. Shao *et al.* [17] presented an improved PSO algorithm to improve the solution's quality. In [18], an improved algorithm is used to overcome the problems of the slow convergence speed and low search efficiency of the ACO algorithm. Based on the different types of journals, intelligent algorithms, and planning standards, the current status of trajectory planning of UAVs is reviewed [19].

In a conclusion, the above algorithms can solve the trajectory planning problem coupled by obstacles and threats. However, the convergence of the algorithm and optimality of the better solution are not all satisfactory for practical application. In addition, collision avoidance between unmanned aerial vehicles and the obstacles is another challenging problem in multiple UAVs trajectory planning. Previous studies solve the problem of obstacle avoidance through the gradient method, the nonlinear optimal control, and dynamic programming, but most of them ignore the size of UAV and just consider them as a single point. So how to improve the convergence speed of the algorithm more efficiently, increase the search efficiency of the optimal solution, and avoid the problem of collisions between UAVs and the obstacles are the motivation of this paper. All UAVs need to autonomously find the optimal trajectory under the specified performance index. In particular, this paper aims to research the global path planning problem for multi-UAV from different starting points to the same target point in a known environment. The main contributions of this paper are as follows.

Firstly, an improved algorithm called MACO is proposed by introducing the metropolis criterion into node screening of the basic ACO algorithm to accept the new state and the sub-optimal solution with a certain probability, and it overcomes the problem that the traditional ant colony algorithm is easy to fall into the local optimum too early. Secondly, the collision risk can easily occur in the case of that the flight trajectory is not well planned if the size of the aircraft is considered in the path planning. For the problem, three trajectory correction schemes are designed in this paper to satisfy the energy-saving requirements and minimum task risk constraints of the trajectory planning. And the theoretical derivation and simulation results show that the scheme 3 is optimal. Lastly, some trajectory segments planned by the above methods are discontinuous, which result in increasing the cost of threat and fuel consumption of the trajectory. This paper uses the inscribed circle (IC) smooth method to smooth the modified trajectory segments.

The rest of this paper is organized as follows: In Section II, the trajectory planning problem is described. The detailed descriptions of the MACO algorithm are given in Section III. In Section IV, three trajectory correction schemes and the IC smooth method are presented. In Section V, simulations are conducted to demonstrate the feasibility of

the proposed method. Finally, Section VI illustrates the conclusions.

II. DESCRIPTION OF UAVS TRAJECTORY PLANNING PROBLEM

The trajectory planning problem is to search for an optimal trajectory from the starting point to the destination point, which is composed of some sequence points [20]. Considering a multi-UAV system with size of n ($num = 1, 2, \dots, n$), the sequence point of every UAV at a certain time can be represented by a binary group ($num, location$). Here, $location = (x, y)$ or (x, y, z) represents the position of the UAV. Fig. 1 shows a schematic diagram of the trajectory planning which taking the environment threat, range constraints, and collision avoidance between UAVs. The MACO algorithm is used to generate the feasible initial trajectory, which can effectively avoid the stagnation phenomena. Since the UAV is regarded as a single point in the initial trajectory planning process, the planned trajectory cannot satisfy the requirement of the safe flight. In this paper, the trajectory correction scheme is adopted to solve the collision avoidance between UAVs and the obstacles. Then, the IC smooth method is taken to smooth the produced trajectory which composes of several polygonal lines.

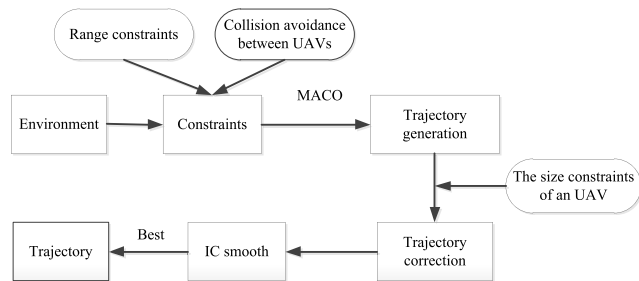


FIGURE 1. The schematic diagram of UAV trajectory planning.

To solve the problem of the global trajectory planning in a known environment, some assumptions are given as follows.

Assumption 1. All UAVs are isomorphic.

Assumption 2. All UAVs have the same constant velocity and safety distance.

A. MODELING OF ENVIRONMENT

The environmental modeling of the trajectory planning can be equivalent to the peaks [5], the cylinders, and cuboids. In the planning area, the UAV's motion in three-dimensional space can be decoupled into two-dimensional plane motion and vertical motion [4], [21], respectively. The mission environment here is two-dimensional plane.

Planning area of UAVs is modeled as a grid map. It is defined as empty (white square grid represents the possible position) or occupied (black square grid represents the infeasible position) according to whether the obstacle boundary is within the grid or not. Grid map is usually used in the trajectory planning with a known environment, and it easily

creates and has low requirement for the sensor accuracy. The mathematical correspondence between the grid number and the position coordinates is calculated by Eq. (1)

$$\begin{cases} x = a^* (\text{mod}(i, n) - 0.5) \\ y = a^* (0.5 + n - \text{ceil}(i/n)) \end{cases} \quad (1)$$

here a denotes the scale of the grid, and it is determined by the size of the UAV. i denotes the number of grids. n denotes the number of rows in the grid. mod denotes the remainder function. ceil denotes the rightward rounding function.

B. RANGE CONSTRAINTS

1) PATH SEGMENT LENGTH

Before changing the flight attitude, the distance traversed in a straight line by the UAV must be greater than a preset minimum value, expressing by $l_i \geq l_{\min}$, $i = 1, \dots, n$. Here l_i represents the distance traversed in a straight line by the i th UAV and $l_{\min}$ denotes the minimum length of the path segment.

2) DISTANCE CONSTRAINTS

The length of the trajectory must be less than or equal to a preset maximum value, which is limited by the amount of fuel and flight time. Let $L_{\max}$ be the maximum length of the trajectory, where $\sum_i l_i \leq L_{\max}$.

C. COLLISION AVOIDANCE CONSTRAINTS

The collision avoidance problem is usually divided into two types: collision avoidance between UAVs and the obstacles and between UAVs.

1) COLLISION AVOIDANCE BETWEEN UAVS AND THE OBSTACLES

In the planning area, the obstacles such as the environment are rasterized and marked as the black squares. The UAVs avoid collisions with the obstacles by not overflying the cells of the grid occupied by obstacles (the black cells of the grid in Fig. 2 represent the obstacles).

2) COLLISION AVOIDANCE BETWEEN UAVS

In order to avoid the collisions between UAVs, the minimum distance between the i th UAV and j th UAV should be not less than the safe distance, that is $d_{ij} \geq d_{\min}$. The collision probability of any two UAVs can be calculated by equation (2)

$$p_{ij} = \frac{|d_{ij} - d_{\min}|}{d_{\min}} \quad (2)$$

d_{ij} denotes the distance between the i th UAV and j th UAV, and $d_{\min}$ denotes the safe distance between the i th UAV and j th UAV.

D. OBJECTIVE FUNCTION

Considering a multi-UAV system with size of n , and regarding the evaluation of a candidate trajectory, the objective

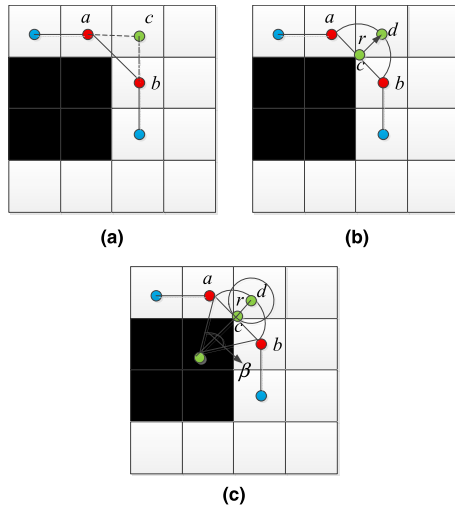


FIGURE 2. Three trajectory correction schemes. (a) The schematic diagram of correction scheme 1 of trajectory. (b) The schematic diagram of correction scheme 2 of trajectory. (c) The schematic diagram of correction scheme 3 of trajectory.

function can be built using Eq. (3)

$$J = \sum_{i=1}^n \mu_i l_i \quad (3)$$

l_i represents the optimal trajectory length of the i th UAV which can be given by adding all the distances of trajectory segments in the planned trajectory, and μ_i denotes the relative weight coefficient. The primary mission in the next part is to find the optimal or the suboptimal solutions by the objective function.

III. IMPLEMENTATION OF THE PROPOSED ALGORITHM

This section first introduces the metropolis criterion, and then proposes an improved MACO algorithm that is used in the paper.

A. METROPOLIS CRITERION

The metropolis criterion inspired from the simulated annealing algorithm is an effective method to accept the new state and the suboptimal solution with a certain probability, and it can avoid falling into the local optimal solution too early [5]. Algorithm 1 summarizes the procedure of the metropolis criterion.

k represents the number of iterations. m denotes the sequence numbers of ants. pl denotes the length of current trajectory. min_kl denotes the optimal solution. n denotes the counter. T denotes the current temperature parameter. $rand$ represents a random number. $minlength_ant(n)$ represents the sequence numbers of ants corresponding to the optimal path at the $minlength_k(n) - th$ iteration.

B. MACO ALGORITHM

ACO algorithm [20]–[23] is a kind of intelligent meta-heuristic algorithm inspired by the foraging behavior of ants.

Algorithm 1 The Metropolis Criterion

```

1 Initialize:  $k, m, pl = 0, min\_kl = \infty, n = 0$ ;
2 While not stopping criterion do
3    $n = n + 1$ ;
4   Generate  $pl$ ;
5   if  $\exp(-(pl - min\_kl)/T) > rand$ 
6      $minlength\_k(n) = k$ ;
7      $minlength\_ant(n) = m$ ;
8      $min\_kl = pl$ ;
9   else
10     $minlength\_k(n) = k$ ;
11     $minlength\_ant(n) = m - 1$ ;
12     $min\_kl = min\_kl$ ;
13  end
14 end
15 Return the best solution  $min\_kl$ ;
```

According to the pheromones deposited by ants, other ants can move to the trajectory with the higher pheromone concentration than before. Finally, the ants are able to find an optimal trajectory between the nest and food by a remarkably effective way.

The ACO algorithm has the advantages of the parallelism, the strong robustness, and higher efficiency, but it easily falls into stagnation [24]. In order to solve the problem, the metropolis criterion is introduced into the selection mechanism of the ACO algorithm. Therefore, an improved MACO algorithm is presented to generate the initial trajectory. And the algorithm can simultaneously perform trajectory planning for different populations. Each ant corresponds to one trajectory, and all candidate trajectories of each UAV constitute an ant population. The ants in each population have position, navigation, perception and communication, and interaction capabilities the same as the UAV platform [25]. Algorithm 2 defines the trajectory generation algorithm using the MACO.

Algorithm 2 The MACO Algorithm

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1 Initialize: The number of ants, the maximum number of iterations, the current number of iterations, the pheromone heuristic factor, the expected heuristic factor, and the threshold;
2 Calculate the probability that ant  $k$  selects the next node  $j$  by equation (4);
3 Based on the metropolis criteria, the suboptimal solutions that satisfied the constraints are accepted;
4 The pheromone is updated by equation (5);
5 If the maximum number of iterations is reached or the end condition is satisfied, output the optimal trajectory length, otherwise, go to 1;
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The state transition rule is the most basic part of the MACO algorithm, which is used to select the next grid. Equation (4) can obtain the state transition probability, which is deduced

by the pheromone concentration and heuristic value between the current node and the next alternative node.

$$p_{ij}^{(k)}(t) = \begin{cases} \frac{\tau_{ij}^{\alpha}(t) \eta_{ij}^{\beta}(t)}{\sum_{j \in allowed_k} \tau_{ij}^{\alpha}(t) \eta_{ij}^{\beta}(t)} & \text{if } j \in allowed_k \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

α and β denote the pheromone heuristic factor and expected heuristic factor, respectively. α reflects the importance of the pheromone concentration on the trajectory search, and β reflects the importance of the heuristic information on trajectory. $\tau_{ij}(t)$ represents the pheromone concentration between node i and node j at time t , and $\eta_{ij}(t)$ denotes the heuristic value between node i and node j at time t . $allowed_k$ is the set of candidate nodes. The heuristic value can be calculated by $\eta_{ij}(t) = \frac{1}{d_{ij}}$, where d_{ij} denotes the Euclidean distance between node i and node j .

The pheromone concentrations of different grids directly affect the motion direction of ants. The pheromone update mechanism can be divided into the local pheromone update mechanism and global pheromone update mechanism. The local pheromone update mechanism can effectively avoid excessively searching for a certain area, and the global pheromone update mechanism can ensure the continuous coverage of the entire mission environment [4]. Only the global pheromone is updated in MACO algorithm. The pheromone at time $t + 1$ can be obtained by equation (5),

$$\tau_{ij}(t + 1) = (1 - \rho) \tau_{ij}(t) + \Delta \tau_{ij}(t) \quad (5)$$

The pheromone volatility coefficient ρ directly affects the global search ability and convergence speed of the algorithm. The bigger the ρ is, the stronger randomness and global search capability of the algorithm. The pheromone increment $\Delta \tau_{ij}(t)$ at time t can be expressed as

$$\Delta \tau_{ij}(t) = \begin{cases} Q/L_k(t), & \text{the } k\text{th ant from node } k \text{ to node } j \\ 0, & \text{else} \end{cases} \quad (6)$$

let Q be the strength of the pheromone, and L_k be the length of the trajectory taken by the k th ant passes.

IV. THE TRAJECTORY OPTIMIZATION PROCESS

In this section, three trajectory correction schemes and the IC smooth method are given to further optimize the initial trajectory.

A. THREE TRAJECTORY CORRECTION SCHEMES

When the size of the aircraft is considered in the trajectory planning, the planned trajectory may collide with the obstacles. For safe flying, this part designs three schemes to avoid collisions between unmanned aerial vehicles and the obstacles, and three trajectory correction schemes are shown in Fig. 2.

1) CORRECTION SCHEME 1 OF TRAJECTORY

Fig. 2a is a schematic diagram of correcting a local trajectory segment by the scheme 1. A new node $c(x_c, y_c)$ is added in the trajectory segment ab which collides with the obstacles. c is at the perpendicular bisector of the trajectory segment ab generated by node $a(x_a, y_a)$ and node $b(x_b, y_b)$, and the coordinate of c can be described as $x_c = x_b, y_c = y_a$. So the original trajectory ab is converted into $ac + cb$ by the correction scheme 1.

2) CORRECTION SCHEME 2 OF TRAJECTORY

In Fig. 2b, there is a risk of collision between the trajectory segment ab and the obstacle, $a(x_a, y_a)$ denotes the previous node, and $b(x_b, y_b)$ denotes the latter node. Here, points a and b are symmetric about point $c(x_c, y_c)$, there $x_c = \frac{x_a + x_b}{2}$ and $y_c = \frac{y_a + y_b}{2}$. Let point c be the center of the circle, and $r_c = \frac{1}{2}|ab|$ be radius. The position of the point d is determined by the position of the obstacle and the direction of the flight speed of UAV, and $\widehat{adb}$ generated by the points a, d , and b does not collide with the obstacles. So the original trajectory ab is converted into $\widehat{adb}$ by the correction scheme 2.

3) CORRECTION SCHEME 3 OF TRAJECTORY

In Fig. 2c, there is a risk of collision between the trajectory segment ab and the obstacle, and the points $a(x_a, y_a)$ and $b(x_b, y_b)$ are symmetric about the point $c(x_c, y_c)$, here, $x_c = \frac{x_a + x_b}{2}$ and $y_c = \frac{y_a + y_b}{2}$. From the geometric principle, the line ab generated by the points a and b can be expressed as l_{ab} . $d(x_d, y_d)$ on the perpendicular bisector of the segment ab can be found. r_d denotes a safety distance from the point d to l_{ab} . In simulation experiment, let $r_d = 50$ m be the safety distance. That is, let the point d be the center of the circle o_d , and the circle with the radius r_d is tangent to l_{ab} . And there are two points on the perpendicular bisector of the segment ab and a point d in the safe area. Thus, $\widehat{adb}$ is generated by the points a, d , and b . The original trajectory ab is converted into $\widehat{adb}$ by the correction scheme 3.

The point d is calculated as follows:

From the points $a(x_a, y_a)$ and $b(x_b, y_b)$, k_1 denotes the slope of l_{ab} , and $-\frac{1}{k_1}$ denotes the slope of l_{cd} , that is, $l_{cd}: y = -\frac{1}{k_1}x + b$. The point c is on l_{cd} .

$$y_c = -\frac{1}{k_1}x_c + b \quad (7)$$

Equation (7) can be changed to $b = y_c + \frac{1}{k_1}x_c$, then

$$l_{cd}: y = -\frac{1}{k_1}x + y_c + \frac{1}{k_1}x_c = -\frac{1}{k_1}(x - x_c) + y_c \quad (8)$$

The point d is on l_{cd} .

$$y_d = -\frac{1}{k_1}x_d + y_c + \frac{1}{k_1}x_c = -\frac{1}{k_1}(x_d - x_c) + y_c \quad (9)$$

$r_d = 0.5$ is obtained by the distance between two points.

$$(x_d - x_c)^2 + (y_d - y_c)^2 = r_d^2 \quad (10)$$

$$(x_d - x_c)^2 + \left(-\frac{1}{k_1}(x_d - x_c) + y_c - y_c\right)^2 = r_d^2 \quad (11)$$

$$\begin{cases} x_d = \sqrt{\frac{r_d^2}{1 + \frac{1}{k_1^2}}} + x_c \\ y_d = -\frac{1}{k_1}x_d + y_c + \frac{1}{k_1}x_c = -\frac{1}{k_1}\sqrt{\frac{r_d^2}{1 + \frac{1}{k_1^2}}} + y_c \end{cases} \quad (12)$$

The points a , d , and b are all on the circle o , (x_o, y_o) represents the circle center coordinate, and the radius of the circle is denoted as r_o , then

$$\begin{cases} (x_a - x_o)^2 + (y_a - y_o)^2 = r_o^2 \\ (x_b - x_o)^2 + (y_b - y_o)^2 = r_o^2 \\ (x_d - x_o)^2 + (y_d - y_o)^2 = r_o^2 \end{cases} \quad (13)$$

$$\begin{cases} x_0(2x_b - 2x_a) + y_0(2y_b - 2y_a) = x_b^2 + y_b^2 - x_a^2 - y_a^2 \\ x_0(2x_d - 2x_a) + y_0(2y_d - 2y_a) = x_d^2 + y_d^2 - x_a^2 - y_a^2 \end{cases} \quad (14)$$

The circle center coordinate can be computed as (15) and (16), as shown at the bottom of this page.

The radius r_o of the circle is computed as

$$r_o = \sqrt{(x_a - x_o)^2 + (y_a - y_o)^2} \quad (17)$$

The arc angle is given by

$$\beta = 2 \arcsin \frac{1}{\sqrt{2} \times r_o} \quad (18)$$

Then, let $\widehat{adb} = \beta r_o$ be the arc length, where β denotes the radian value.

B. INSCRIBED CIRCLE(IC)TRAJECTORY SMOOTH

Fig. 3 displays some trajectory segments in simulation. Here, two trajectory segments, consisting of the node A , the node B , and the node C , occur sharp turns at B . To save the cost of the trajectory planning, the UAV should avoid sharp turns as much as possible in actual flight. Three methods such as the Bezier curve, the B-splines, and the Cubic spline [9], [26], [27] are generally used to smooth the polygonal trajectory in trajectory planning. Inspired by the above methods, this paper used an inscribed circle smooth method to solve this problem.

The inscribed circle can be found in the triangle which consists of A , B , and C . The radius and center of the inscribed circle are r and o , respectively. The inscribed circle is tangent to the trajectory segment AB , and the tangent point is D .

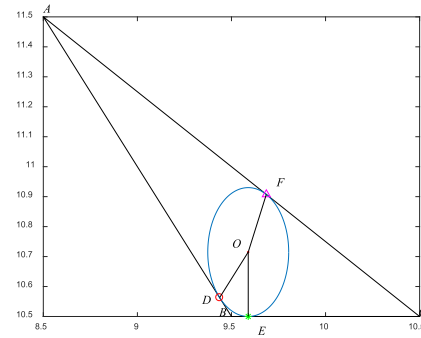


FIGURE 3. The schematic diagram of the IC smooth.

In addition, the inscribed circle is tangent to the trajectory segment BC , and the tangent point is E . The original trajectory segments $AB + BC$ are corrected to $AD + \widehat{DE} + EC$ by the IC smooth method. The length of the original trajectory which consists of A , B , and C is $|AB| + |BC| = 2.4142 \text{ units}$ (241.42 m). The length of the smooth trajectory is $|AD| + |\widehat{DE}| + |EC| = 2.4050 \text{ units}$

(240.50 m). It can be seen that the modified trajectory are smoother than the original trajectory and the trajectory length is shortened 0.29 m.

V. SIMULATION EXPERIMENT

In this section, simulations and comparisons are performed in MATLAB R2016b to verify the performance and validity of the proposed method. The configuration of the computer is as follows:

1. System: windows 7 enterprise 64 bit.
2. Processor: Intel(R) Core(TM) i5-4460 CPU 3.20GHZ.
3. Memory: 4.0GB.

A. SINGLE UAV PATH PLANING

The mission area is discretized to a grid map of 20×20 cells, each one with 200 m. The parameters of the trajectory planning problem are listed in Table 1.

TABLE 1. The maco algorithm parameters setting.

Parameter	Value
The number of ants	30
The maximum number of iterations	200
The pheromone volatility value	0.3
The pheromone heuristic factor	1
The expected heuristic factor	5
The strength of the pheromone	5

$$y_0 = \frac{(x_d^2 + y_d^2 - x_a^2 - y_a^2)(2x_b - 2x_a) - (x_b^2 + y_b^2 - x_a^2 - y_a^2)(2x_d - 2x_a)}{(2y_d - 2y_a)(2x_b - 2x_a) + (2y_b - 2y_a)(2x_d - 2x_a)} \quad (15)$$

$$x_o = \frac{x_b^2 + y_b^2 - x_a^2 - y_a^2}{2x_b - 2x_a} - \frac{2y_b - 2y_a}{2x_b - 2x_a} y_o \quad (16)$$

1) EXPERIMENTAL COMPARISON

The performance of the MACO algorithm is evaluated by independently repeating 20 times. In the same environment and parameters, the comparison results of the MACO algorithm and the algorithm [18] with respect to the trajectory planning are given in Table 2. These results are measured in terms of the worst value (Worst), the best value (Best), and the average value (Mean). Fig. 4a and Fig. 4b show the tendency of the optimal trajectory length and running time of the two algorithms. In Fig. 4c, the red line is the optimal trajectory of the MACO algorithm, and the blue line is the optimal trajectory of the algorithm in [18].

TABLE 2. Performance comparison of two algorithms.

No.	Algorithm in [18]		MACO	
	The optimal path length	Time	The optimal path length	Time
1	28.6274	42.7128	28.6274	17.2318
2	29.2132	50.4460	28.6274	13.5688
3	29.2132	46.3469	28.6274	14.5665
4	29.2132	47.7129	28.6274	13.9964
5	29.2132	49.6069	28.6274	14.4361
6	28.6274	51.9698	28.6274	16.3086
7	29.2132	53.6251	28.6274	14.2976
8	28.6274	48.8424	28.6274	15.4541
9	29.2132	38.7494	28.6274	14.1483
10	29.2132	44.9374	28.6274	15.7294
11	28.6274	41.4700	29.2132	17.0624
12	28.6274	57.3220	28.6274	16.3258
13	28.6274	50.3456	28.6274	16.2955
14	28.6274	41.9611	28.6274	15.6801
15	29.2132	52.0422	28.6274	14.6112
16	29.2132	40.4981	28.6274	15.3140
17	28.6274	46.3022	28.6274	15.5714
18	28.6274	45.3030	28.6274	15.5291
19	28.6274	52.1801	28.6274	15.0247
20	29.2132	50.4252	28.6274	15.3861
Worst	29.2132	57.3220	29.2132	17.2318
Best	28.6274	38.7494	28.6274	13.5688
Mean	28.9203	47.6400	28.6567	15.3269

In Fig. 4a, all optimal trajectory length values obtained in the MACO algorithm are almost less than the results in [18]. The best value and the worst value of two algorithms are same in Table 2, but the average value of the MACO algorithm is less than the average value of the algorithm in [18]. In Fig. 4b, the running time of the MACO algorithm to search for the optimal trajectory is always shorter than the running time obtained in [18]. In Fig. 4c, the optimal trajectory in the MACO algorithm is better than the result obtained in [18]. That is, the value of cost function corresponding to the improved algorithm is superior to the algorithm in [18]. And the trajectory generated by the MACO algorithm is more feasible than the algorithm in [18].

2) TRAJECTORY PLANNED BY THE MACO ALGORITHM

In this section, the obstacles are randomly generated. The starting point is (0.5, 19.5) and target point is (19.5, 0.5). The maximum number of iterations is set as 200. The above data take the side length of a grid as its unit length. Fig. 5a and

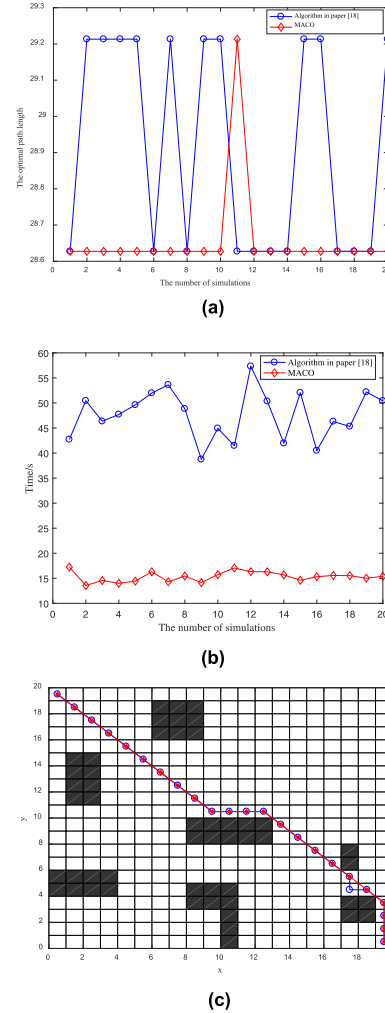


FIGURE 4. Comparison results of two algorithms in trajectory planning. (a) The convergence curve of the optimal trajectory length. (b) The running time of two algorithms. (c) The trajectory planning in two algorithms.

Fig. 5b show the results of the initial trajectory planning by the MACO algorithm. Fig. 5c depicts the comparison results of the optimal trajectory length after adding the metropolis criterion.

In Fig. 5a, the length of the initial optimal trajectory is 2862.74 m (28.6274 units). As shown in Fig. 5b, the MACO algorithm can achieve an optimal trajectory after first iteration, and the average trajectory length is relatively stable after 20 iterations. In Fig. 5c, the black curve is generally below the blue curve, that is to say, the improved algorithm of this paper can quickly search for an optimal trajectory. Simulation experiments illustrate that the MACO algorithm not only can obtain a stable solution, but avoid falling into stagnation effectively.

3) TRAJECTORY CORRECTION BASED ON THE SCHEME 3

In Fig. 5a, there is a risk of collision between obstacle 1# and the trajectory segment generated by the 13th node and the 14th node, and the other cases occur in the obstacle 2#

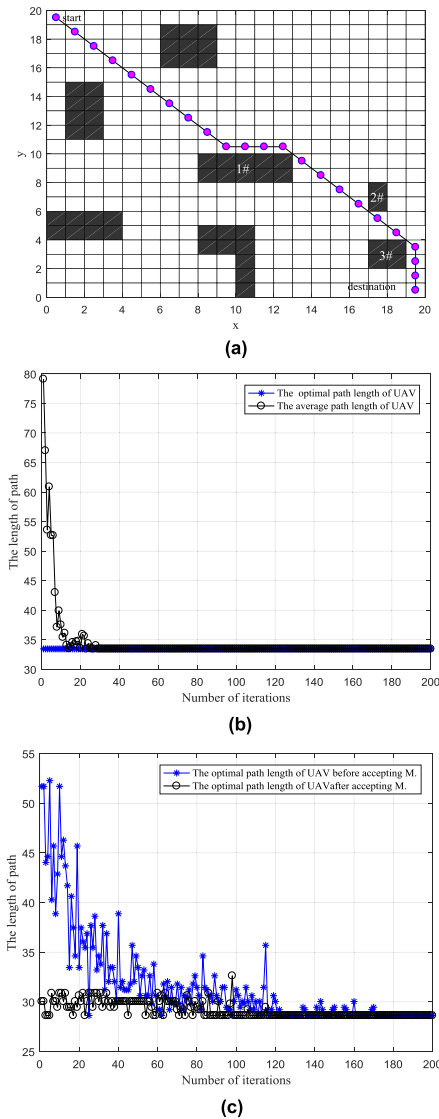


FIGURE 5. The trajectory planned by the MACO algorithm. (a) Trajectory for single UAV. (b) The convergence curve of the trajectory length. (c) The comparison results of the optimal trajectory length after adding the metropolis criterion.

TABLE 3. The optimal path length in different methods.

Method	The optimal path length (units)
MACO	28.6274
Scheme 1	30.3848
Scheme 2	31.0419
Scheme 3	29.9240

and obstacle 3# similarly. To solve the collision problem, the initial trajectory is further optimized by three correction schemes, and the results are shown in Table 3. The results show that all schemes can make a little sacrifice in the trajectory length, so as to effectively solve the collision avoidance problem. In particular, the corrected trajectory length of scheme 3 is 46.08 m (0.4608 units) shorter than the scheme 1 and 111.79 m (1.1179 units) shorter than the scheme 2.

This illustrates that the scheme 3 is superior to the scheme 1 and scheme 2. Therefore, the initial trajectory is corrected only using the scheme 3 in the trajectory planning.

Base on the scheme 3, as shown in Fig.6, three new '*' nodes are added to between the 13th node and the 14th node, the 17th and the 18th node, and the 19th and the 20th node, respectively. It is obvious that the corrected trajectory can successfully avoid all obstacles.

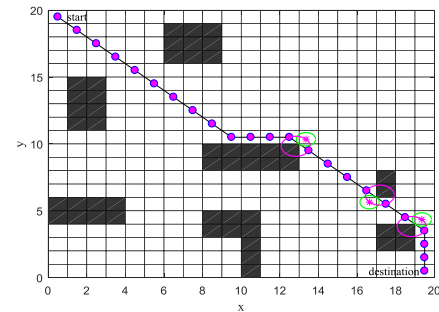


FIGURE 6. Trajectory correction based on the scheme 3.

B. MULTI-UAV (3 UAVS) PATH PLANNING

In simulation, the initial positions of UAV1, UAV2, and UAV3 correspond to the serial numbers 1, 21, and 161 of the grids, respectively, and the final position of three UAVs corresponds to the serial number 400 of the grid. In order to verify the stableness and generalization of the MACO algorithm, the experiments are performed in two different planning scenarios. The simulation results are listed as follows.

1) RESULTS OF THE PATH PLANNING IN SCENARIO 1

Fig. 7a-7d show the results of trajectory planning problem using the MACO algorithm. The total running time in the simulation is 38.799 seconds. It can be seen from Fig. 7a-7c that the improved algorithm can quickly search for the optimal trajectory. As the number of iterations increases, the average path length converges to a certain value. In Fig. 7d, the optimal initial trajectory length of UAV1, UAV2, and UAV3 are 2862.74 m, 2821.32 m, and 2521.32 m, respectively. The corresponding serial numbers of the optimal initial trajectory of UAV1, UAV2, and UAV3 are [1, 22, 43, 64, 85, 106, 127, 148, 169, 190, 191, 192, 193, 214, 235, 256, 277, 298, 319, 340, 360, 380, 400], [21, 42, 62, 83, 104, 125, 146, 167, 188, 208, 228, 249, 250, 251, 272, 293, 314, 335, 356, 377, 398, 399, 400], and [161, 182, 203, 224, 205, 226, 247, 268, 289, 270, 291, 312, 333, 354, 375, 396, 397, 398, 399, 400], respectively. Since the speed of three UAVs is same, there is no collision among three initial trajectories generated by the improved algorithm. Experimental results show that the MACO algorithm significantly reduces the probability of falling into local optimum, and it also improves the ability of finding the optimal trajectory.

Fig. 8a, 8b, and 8c show the optimal trajectory length for three UAVs in different algorithms. The black line represents the optimal trajectory length of the MACO algorithm,

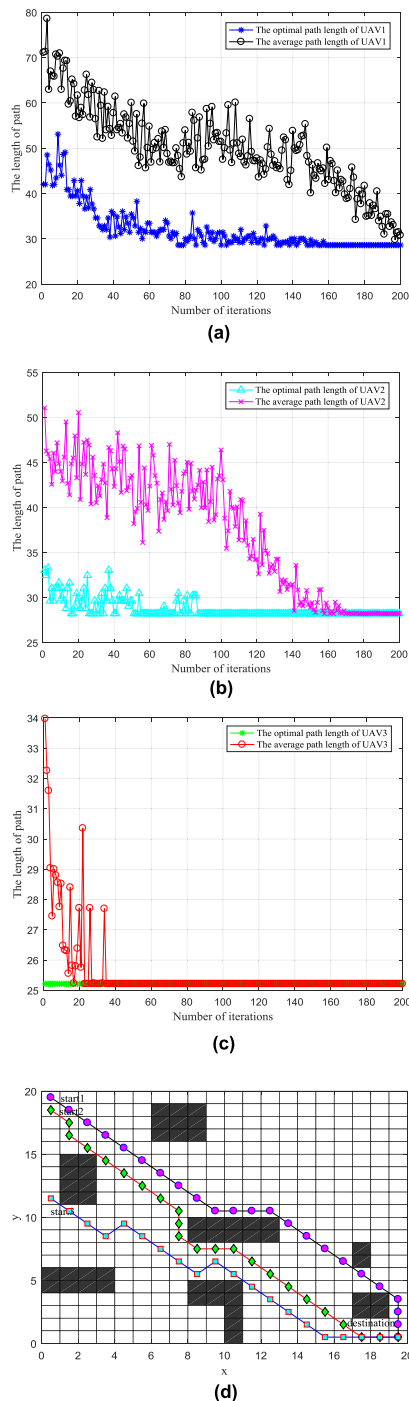


FIGURE 7. The results of the trajectory planning in scenario 1. (a) Convergence curve of the path length for UAV1. (b) Convergence curve of the path length for UAV2. (c) Convergence curve of the path length for UAV3. (d) Trajectory planning by the MACO algorithm.

and the blue line is the optimal trajectory length of the ACO algorithm. These figures show that the optimal initial path length of the MACO algorithm is relatively stable, and the improved algorithm has faster convergence. Therefore, the MACO algorithm can achieve high performance and stability compared with the traditional ACO algorithm.

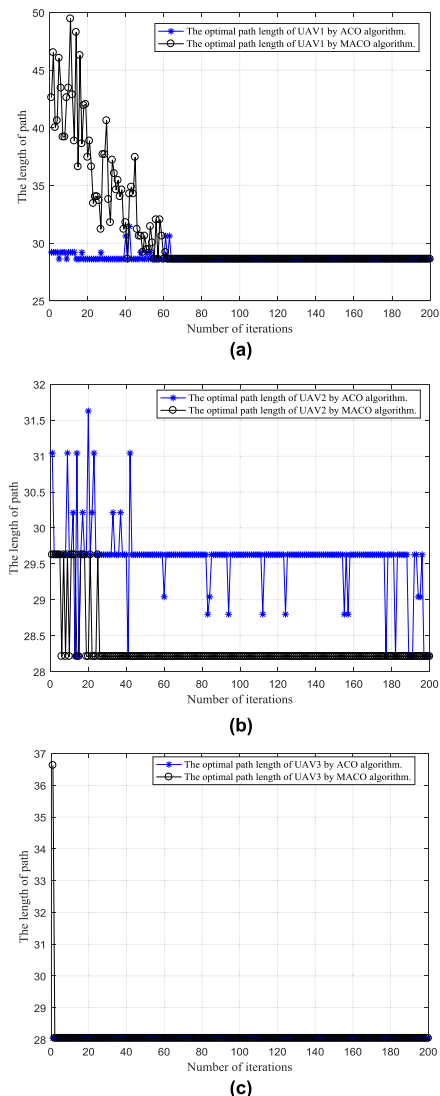


FIGURE 8. Comparison results of the optimal path length for three UAVs in scenario 1. (a) Comparison results of the optimal path length for UAV1. (b) Comparison results of the optimal path length for UAV2. (c) Comparison results of the optimal path length for UAV3.

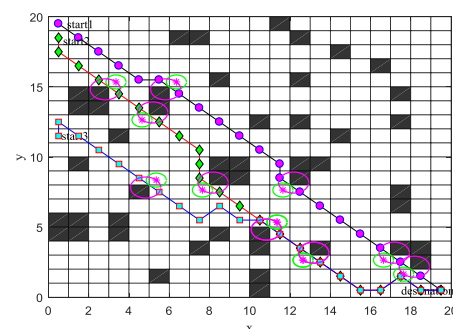


FIGURE 9. Trajectory correction based on the scheme 3.

2) RESULTS OF THE PATH PLANNING IN SCENARIO 2

Considering the geometric size of the aircraft, it can be seen that three initial trajectories generated by the MACO algorithm collide with the obstacles, and this problem can be well

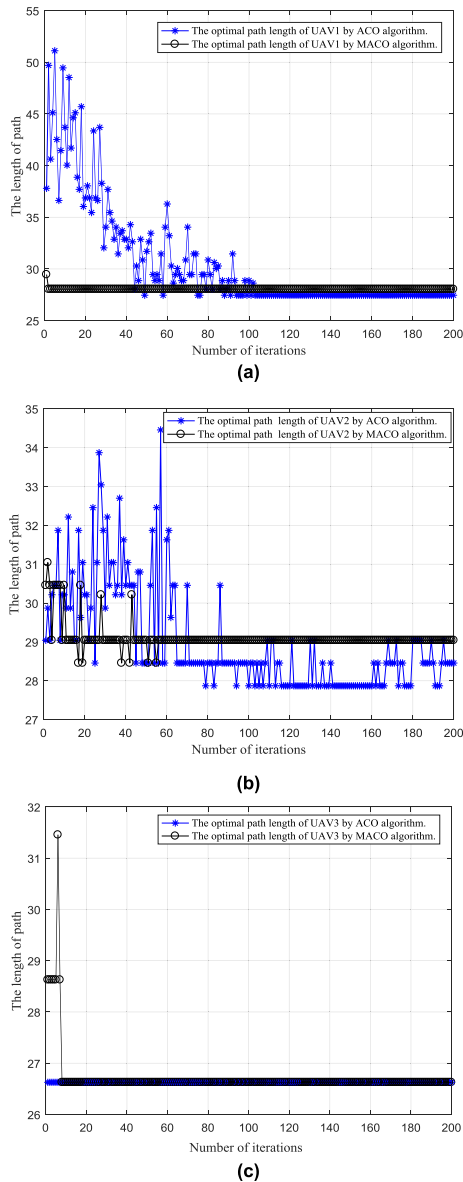


FIGURE 10. Comparison results of the optimal path length for three UAVs in scenario 2. (a) Comparison results of the optimal path length for UAV1. (b) Comparison results of the optimal path length for UAV2. (c) Comparison results of the optimal path length for UAV3.

solved using scheme 3. In Fig. 9, ten new ‘*’ nodes are added in the three trajectories, and it is obvious that the modified trajectories can avoid all threat. In Fig. 10a, 10b, and 10c, the optimal initial trajectory length of each UAV is given in detail. Compared with Fig 8, they show that the improved algorithm can still quickly find the optimal trajectory in more complicated scenario 2 and require less iteration than the ACO algorithm. That is, the experimental results show that the MACO algorithm not only can effectively avoid falling into the local optimal, but can achieve faster convergence speed.

Fig. 11a shows the smoothed trajectory of UAV1 in scenario 2. Fig. 11b is a partial view of Fig. 11a. It is obvious that

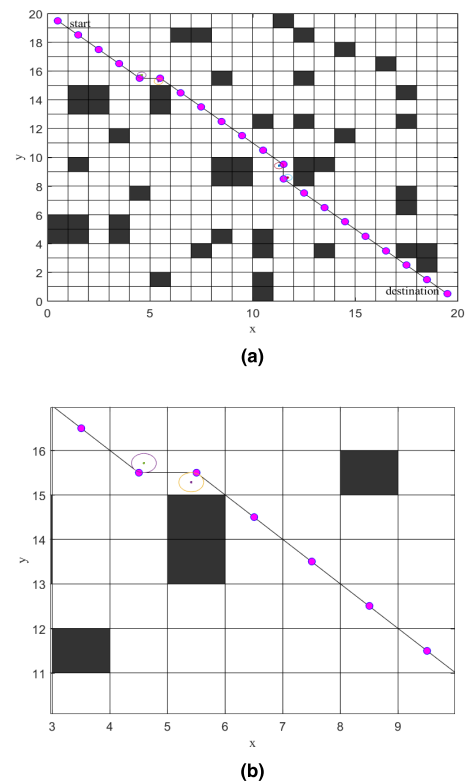


FIGURE 11. Smooth planning by the IC smooth method. (a) Smooth trajectory planning. (b) Partial view of Fig. 11a.

the three trajectory segments consist of four nodes (3.5, 16.5), (4.5, 15.5), (5.5, 15.5), and (6.5, 14.5), and they form a discontinuous trajectory. Based on the IC smooth method, nodes (4.4370, 15.5630) and (4.5891, 15.5) are added in the two trajectory segments, and these trajectory segments compose of nodes (3.5, 16.5), (4.5, 15.5), and (5.5, 15.5). Let (4.5891, 15.7150) be the center of the circle, and 0.2150 be the radius. Meanwhile, nodes (5.4109, 15.5) and (5.5630, 15.4370) are added in the two trajectory segments which consist of nodes (4.5, 15.5), (5.5, 15.5), and (6.5, 14.5). Let (5.4109, 15.2850) be the center of the circle, and 0.2150 be the radius. The above data take the side length of a grid as its unit length. Similarly, the other trajectory segments are smoothed by this method. It can be seen from Fig. 11b that the trajectory is more smoothness than before. In a word, the smooth trajectory generated by the IC smooth method can effectively avoid sharp turns and reduce the randomness of the result.

VI. CONCLUSION

In this study, the MACO algorithm is proposed to solve the problems of the local optimal solution and slow convergence speed in the ACO algorithm. And it can successfully search for the optimal trajectory from the initial point to the target point in a known environment. Compared with the algorithm in [18], the MACO algorithm can quickly find the optimal trajectory in the same mission scenario. Compared with the

ACO algorithm, the proposed algorithm does not require additional iterations to get the optimal trajectory in more complicated scenario 2, and the probability of falling into local optimum of the proposed algorithm is significantly reduced. Considering size constraint of UAVs, the initial trajectory may collide with the obstacles. Three trajectory correction schemes are utilized to further optimize the initial trajectory, and the modified trajectory can avoid every obstacle successfully. Furthermore, this paper adopts the inscribed circle smooth scheme to smooth the modified trajectory. The simulation results show that the proposed method can get higher quality trajectories efficiently. Finally, the application of the proposed method should be further explored in the real-time trajectory planning.

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